VF Asia Footwear Laboratory Testing Manual

Test Method for Lace Tip Retention	Test Method #: VF- B24 REF #: TM-R-049 Page 1 of 2
Date Issued: May,2019 Revision #: 1 Reason for revision: Rectify the unit of measure from kg to kgf for material strength	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the attachment strength of a tag to a shoelace.

REFERENCES: Method PM175

EQUIPMENT AND MATERIAL:

1. Tensile testing Machine with clamping jaws or with pneumatic grip.
2. A jig for holding the test shoe lace tag consisting of a flat metal plate of thickness approximately 3mm with a "V" shaped slot cut in it which has:
    - Vertical edges
    - A width at the top of the "V" approximately 3mm
    - A length of approximately 35mm
3. Lace with tip.

SAMPLE PREPARATION:

Condition the specimens at 23 ± 2 ℃ temperature and 55 ± 10% relative humidity for 24 hours prior to test.

REPETITONS: Test six (6) samples

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Clamp the tag of a test shoelace in the jig by sliding the lace into the slot so that the base of the tag is adjacent to the metal plate.

2. Attach the jig to one of the jaws of the tensile testing machine so that the tag is aligned with the axis of the tensile testing machine.

3. Clamp the main body of the test lace in the other jaw of the tensile testing machine so that the lace is just taut and its longitudinal axis is 90° to the clamping edges.

4. Operate the tensile testing machine with a jaw separation rate of 100 ± 10 mm/min until the specimen fails.

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5. Record the force at break in kgf

6. Reject any results and test fresh specimens where:

- The tag slipped through the “V” shaped slot without being removed from the lace.
- The tag slipped through the “V” shaped slot and was damaged by scraping.
- The tag buckled at its base.

7. Repeat step #1 through step # 5 until six valid results have been obtained.
8. Calculate the average of the six maximum forces.

REPORT:

1. Report the average of the average of six maximum forces in kgf.
2. Report lace description, the average force at break, any deviations from this procedure, and any deviations from this procedure.